

Certified analytical bounds for hypersonic reachable sets

A two-page summary of a preliminary PhD research proposal

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Full proposal: research.ekdynamics.science/proposal.pdf

The result. This is a problem in nonlinear analysis: a singularly perturbed family of differential inclusions in which compactness fails along the time axis, because the motion carrying the important state transitions occupies a vanishing time — so it disappears from the ordinary limits of the family while continuing to decide where the trajectories go. For a model keeping exactly that phenomenon I can prove

$$\mathcal{R}_\varepsilon(T) \subseteq \underbrace{\left[\mathcal{R}_\mathcal{H}([T - \varrho, T + \varrho]) + \Lambda \times \{0\} \right]}_{\text{skeleton, bubbles and glide}} \oplus \underbrace{\mathbb{B}(\varrho)}_{\text{residual}},$$

where Λ lifts a state of the averaged mode to the fast fibre it stands for, and the residual is a sum of three terms — the drift dropped over the passes, the error of averaging the slow motion, and the air the ballistic arcs discard — each bounded in closed form by one estimate, with every constant explicit in the data and free of ε . A single amplitude threshold sets the balance between the three, and at its best choice $\varrho = O(\varepsilon^{1/4})$; I claim the order and not its sharpness. The inclusion is uniform over the controls and over the initial set, and holds for each member of a bounded family of admissible fields, the uncertainty being quantified over at the end rather than allowed to vary during a flight. The proof is two pages and elementary; the statement, the hypotheses and the proof are in the proposal.

The problem. The structural phenomenon is concentration, and the language for it is a profile decomposition: a low-dimensional *skeleton*, one *bubble* per pass, and a *residual*. In the theory of nonlinear dispersive equations such a decomposition comes from iterating the dichotomy of the concentration-compactness principle [1, 2], and what it delivers is structure without a constant — by contradiction. The proposal is that here the structure can be had quantitatively, because the family carries Lyapunov functionals: the dissipation is monotone with total variation bounded uniformly in the parameter, so the number of profiles above any fixed size is bounded in advance rather than extracted. That count is proved. The decomposition it would organise is not, and the theorem above establishes none — it compares a trajectory with a trajectory of the limit system, and no profile appears in it. What I think is interesting is the pair of facts that follow: the estimate is the one any decomposition would have to reproduce, and the concentration structure turns out to be reachable without the decomposition machinery at all, by an exact integral identity and a budget. Whether the machinery buys anything beyond that is the first question of the programme.

The example that supplies the problem is hypersonic atmospheric entry, where the solution set is the footprint a vehicle can reach and the parameter is the ratio of the atmospheric scale height to the planetary radius. It is the example and the test, not the subject. Computing the set exactly means a Hamilton–Jacobi–Isaacs equation on a grid, exponential in the state dimension and capped in practice near four or five states [3], while a rigid-body entry vehicle carries thirteen. That count is the most legible way to say what the decomposition earns, though what finally binds is the enclosure of the jump map: the bound is a pair of functions exhibited and checked, not a mesh.

The model. The state is (z, v, s) , with z an altitude, v its rate and s a vector of slow coordinates, and

$$\dot{z} = v, \quad \dot{v} = c m^\varepsilon(z) - 1 - \kappa v, \quad \dot{s} = F + m^\varepsilon(z) \Gamma, \quad m^\varepsilon(z) = m(z/\varepsilon),$$

with m positive and strictly decreasing and c a scalar control: a unit-mass particle in the potential Φ_c with $\Phi'_c = 1 - c m^\varepsilon$, one soft wall and uniform gravity above it. The small parameter sits in one place only, the width of the band in which the air is felt; the equations of motion carry none. All of m, κ, F, Γ may be chosen from a bounded family — one choice per flight, so the density is enclosed rather than fixed while the control still ranges over a set at each instant. Letting it vary *during* a flight enlarges the enclosure and is an open problem below: the potential depends on that choice, so a time-varying one moves the well beneath the oscillation and the amplitude stops obeying a closed law. The budget, being a common Lyapunov functional, is indifferent; the amplitude is not.

The model is the entry oscillator rather than a caricature assembled for the occasion, and three regimes of flight are three energies $W = \frac{1}{2}v^2 + \Phi_c(z)$ in one well: the minimum is the equilibrium glide, a small oscillation is the phugoid, a large one is a skip. *The dip is short and the arc is long because the well is asymmetric*, and that asymmetry, read at the scale of the band, is the whole of the concentration. Two Lyapunov functionals do the quantitative work — the amplitude W for the fast motion and the dissipation for the slow — and three exact identities organise it. First, ε sets no shape, only a clock, but the clock is read at amplitude W/ε , which is why a bounded oscillation keeps a period of order one while the part of it spent in the band shrinks like $\varepsilon \log(1/\varepsilon)$. Second, integrating the rate equation across a pass gives the impulse it delivers as twice its entry speed over c , exactly — so a pass takes a vanishing share of the time and essentially all of the effect, and the budget admits only so many such passes. Third, the same rate

equation gives the oscillating part of the density as a derivative, which is what bounds the averaging error: where a periodic argument would invoke a zero mean over a closed orbit, here the primitive is controlled by the velocity excursion, so no orbit need close and neither the control nor the damping need be frozen. The trajectory is cut at its own crossings of $z = 0$ and nowhere else. The limit system $\mathcal{H}$ flows ballistically on the arcs, jumps at $z = 0$ by a map J — the solution set of an inner problem carrying no small parameter — and flows by the averaged field $F + \Gamma/c$ elsewhere. The hypotheses, the constants and the two-page proof are in the proposal.

What makes the proof elementary is that the functionals do all the counting: the number of passes above a fixed size, their summed duration, and the accumulated amplification — which is exponential in each pass’s own impulse and so bounded however many passes there are. One discipline runs throughout: the density is unbounded inside the band as $\varepsilon \rightarrow 0$, and is never bounded pointwise there, only by its impulse — which is exactly what the dissipation functional controls. No compactness argument enters, and that is the point rather than an evasion.

Where the measures enter. The theorem hands the occupation-measure programme of Lasserre, Henrion and collaborators [4] a different problem. $\mathcal{R}_{\mathcal{H}}$ is a reachability problem in $2+d$ coordinates with one guard and a set-valued reset — the hybrid form for which the relaxation is known [5]: the Liouville equation is a linear constraint on a measure, a measure on the guard carries the mass across each pass, and the dual certificate is a pair of functions non-decreasing along the flow and across the reset. The singular limit is discharged by the theorem, and what is left is a problem in a few coordinates. Whether the limit is better taken in this topology or at the level of trajectories as in the proof is open, and the two routes are not known to be equivalent.

What remains. The theorem applies to entry, but its residual is not small at the physical ε . A pass entered at flight-path angle γ_e enters through the ratio $\gamma_e/\sqrt{\varepsilon}$, as in the classical Chapman variables [6], and for the Earth $\sqrt{\varepsilon} \approx 0.034$ is about two degrees while entry angles are a few degrees. So the amplitude at which a pass becomes brief is the amplitude at which the motion being averaged becomes large, and no threshold makes both small at once. Four problems follow.

1. **The decomposition itself.** That every sequence of trajectories splits, along a subsequence, into the skeleton, finitely many profiles obtained by rescaling the passes to their own widths, and a remainder small in a stated norm. The count such a decomposition needs is proved; the decomposition is not.
2. **The statement at a fixed ε .** The orbits near the threshold, which neither approximation serves, must be carried by the unapproximated field. That adds no term to the residual, because nothing is approximated; it costs dimension, over a total duration the budget bounds. What has to be shown is a bound on the growth of that enclosure, explicit in the interval widths — and, with it, the class of time-varying atmospheres for which the amplitude law survives, bounded variation in altitude being what the physics supplies.
3. **Separation of scales, by enclosure.** The attitude and elastic motions do not concentrate, and averaging them is the obvious route and the wrong one: an averaging theorem gives an order and not a bound, and holds through a resonance only outside a set of small measure, which an outer bound cannot discard. They act on the trajectory only through the aerodynamic force, so it is the force that must be enclosed. Two facts are to be established: where there is no air there is no torque, and the attitude has no aerodynamic effect in any case; and where there is air the damping rate is proportional to density times speed, while the energy identity fixes the integral of density against the cube of speed, so the budget that bounds the dissipation also bounds the integrated damping from below.
4. **The companion at the level of measures.** That the limit of the occupation measures satisfies the Liouville equation of the limit system, with the mass crossing each pass carried on the graph of J , is a separate statement from the one above, and it is where the concentration machinery is expected to be needed — for the structure rather than for the constants.

The first and fourth are the analysis; the second and third carry the model to the application. The fourth is what I would most value a conversation about: granting the limit of the measures and the profiles it carries, whether the limiting solution set is contained in the one those profiles generate is a further implication, and the hardest link in the chain. It is the part for which I am looking for an advisor.

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